Literature-Implied Answer: The $p = \infty$ Endpoint of Reinov's Proposition 3.6

Automatic Math Discovery Run `fa_banach_001`

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Source Question

The source paper is O. I. Reinov, *Approximation of operators in Banach spaces*, arXiv:1002.3902 [1]. Proposition 3.6 states that for every $p \in [1, \infty)$ there are separable reflexive Banach spaces X and Y such that

$$\overline{X^* \otimes Y}^{\tau_p} = X^* \widehat{\widehat{\otimes}}_p Y.$$

Remark 3.7 then says that it is seemingly unknown whether this remains true for $p = +\infty$. In the parsed source this is `data/parsed/arxiv_sources/1002.3902/source.tex`, lines 1365–1390.

> **3.6. Proposition.** *For every $p \in [1, +\infty)$ there exist (separable and reflexive) Banach spaces X and Y such that*
>
> $$\overline{X^* \otimes Y}^{\tau_p} = X^* \widehat{\widehat{\otimes}}_p Y.$$
>
> *Proof.* Let X and Y be th separable and reflexive spaces such that the natural mapping $Y^* \widehat{\otimes}_{p'} X \to \mathrm{N}_{p'}(Y, X)$ is not one-to-one. By Proposition 1.1, the dual space to $\mathrm{N}_{p'}(Y, X)$ can be identified with a subspace $\overline{X^* \otimes Y}^{\tau_p}$ of the spaces $\Pi_p(X, Y)$ (see also the proof of Proposition 1.9). Since this subspace is reflexive (Proposition 2.9) and $(X^* \widehat{\widehat{\otimes}}_p Y)^* = \mathrm{N}_{p'}(Y, X)$, we have: $X^* \widehat{\widehat{\otimes}}_p Y = \overline{X^* \otimes Y}^{\tau_p}$. ■
>
> **3.7. Remark.** Seemingly, it is unknown whether Proposition 3.6 is true in the case where $p = +\infty$.

Identification

The endpoint follows from Pitt's theorem. Choose real numbers

$$1 < s < r < \infty, \qquad X = \ell_r, \qquad Y = \ell_s.$$

Then X and Y are separable and reflexive. For $p = \infty$, the absolutely ∞-summing norm is the operator norm, so

$$\Pi_\infty(X, Y) = \mathcal{L}(X, Y).$$

Moreover, τ_∞ is the topology of compact-open convergence.

Finite-rank operators are compact-open dense in $\mathcal{L}(X,Y)$: given $T \in \mathcal{L}(X,Y)$, a compact set $K \subset X$, and $\varepsilon > 0$, choose a finite-rank operator $P : X \to X$ with $\sup_{x\in K} \|Px - x\| < \varepsilon/\|T\|$, and then TP is finite-rank and approximates T uniformly on K.

By Pitt's theorem, every bounded operator $\ell_r \to \ell_s$, $1 < s < r < \infty$, is compact [2]. Since ℓ_s has the approximation property, every compact operator from ℓ_r into ℓ_s is a norm limit of finite-rank operators. Hence

$$\mathcal{L}(\ell_r, \ell_s) = \mathcal{K}(\ell_r, \ell_s) = \overline{\ell_r^* \otimes \ell_s}^{\|\cdot\|}.$$

Combining this with compact-open density gives

$$\overline{X^* \otimes Y}^{\tau_\infty} = \mathcal{L}(X,Y) = X^* \widehat{\widehat{\otimes}}_\infty Y.$$

Thus Proposition 3.6 is true at $p = +\infty$ for the pair $(X,Y) = (\ell_r, \ell_s)$, $1 < s < r < \infty$.

Status and Scope

Status: `literature_implied_answer`. This answers the existence question in Reinov's Remark 3.7. It is not presented as a new theorem beyond the standard Pitt theorem plus the endpoint interpretation of Π_∞ and τ_∞. The supporting literature did not explicitly identify itself as answering Reinov's remark; the implication is agent-identified.

Search Evidence

The run indexes were searched for arXiv:1002.3902, approximation of operators, operator approximation, compact approximation, bounded approximation property, local reflexivity, Pitt, and the $p = \infty$ endpoint terminology. No prior packet or attempt for this source question was found.

References

[1] O. I. Reinov, *Approximation of operators in Banach spaces*, arXiv:1002.3902, 2010.

[2] H. R. Pitt, *A note on bilinear forms*, J. London Math. Soc. 11 (1936), 174–180.